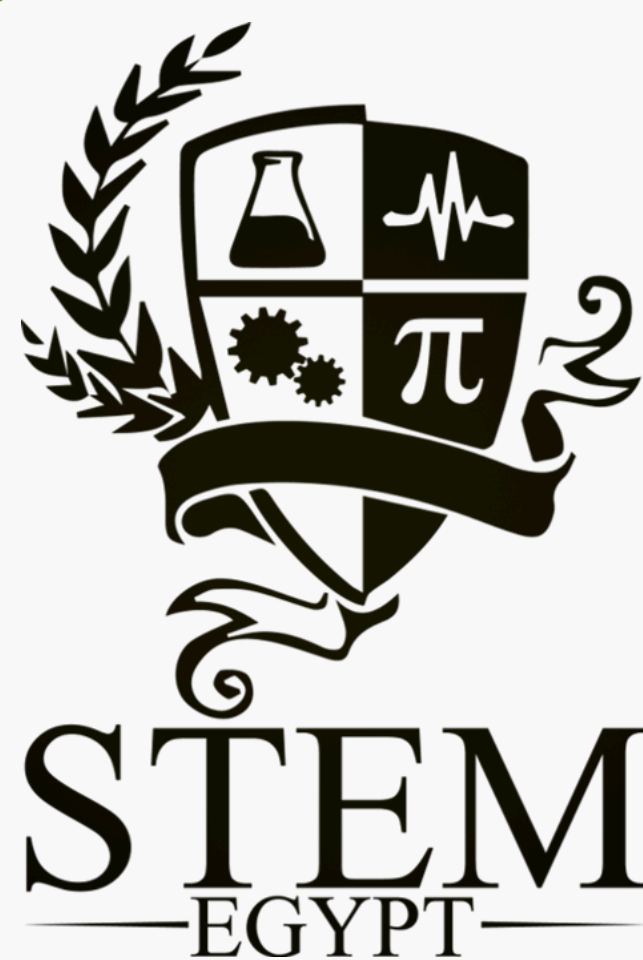




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Abstract

Egypt faces many challenges that hinder its development in some fields, especially the agricultural field. One of these challenges is the decrease in the agricultural and industrial base, which affects the productivity in these fields and affects the Egyptian economy. The purpose of the study is to strengthen the agricultural bases that rely on smart technology and to address problems in the Wadi El Natrun soil, which is low in fertility and highly saline. To solve this problem, it was decided to construct a sawtooth smart greenhouse for two plants, Saffron crocus and lavender. The smart greenhouse relies on a feedback system to maintain the three parameters: temperature, humidity, and soil moisture. The project achieved the design requirements. The prototype volume is 0.125 m³. The construction materials of the project are recycled, and three parameters inside the project are stored in a real-time database. The accuracy of the AI model that detects diseases is above 80%.

Introduction

Egypt, like any country, faces some challenges that hinder its development and weaken its agriculture and industry. One of these challenges is the Industrial and Agricultural Base as Egypt is heavily reliant on the Nile River's water, and many obstacles are reducing water availability. Its agricultural areas haven't exceeded 3.4% of its total area yet, as shown in Figure 1.

Egypt's rapid population growth is a major challenge because it places a heavy burden on the country's limited resources and infrastructure. As a result, the growing population leads to more waste and energy loss.

The specific problem is the intersection of climate change and its strain on economy, particularly evident in the highly saline and degraded soil of Wadi El-Natrun, which prevents traditional crops from growing efficiently while severely compromising plant health and national food security(Yousif & Ahmed, 2019).

To solve these problems, several prior solutions were researched to help understand the problem more correctly. One of the prior solutions is the Ultra-Clima Semi-Closed Greenhouse, a greenhouse that controls the temperature and humidity inside the greenhouse. One of its strengths is advanced pest and disease prevention. One of its weak points is the high cost of installation. Another prior solution is the Spread Smart Vertical Farm, a smart vertical farm in Kyoto, Japan. One of its strengths is water and waste efficiency. One of its weak points is high energy requirements.

Transitioning to the solution, the proposed solution is a smart greenhouse designed specifically for growing lavender and saffron. This system uses a Peltier module to precisely regulate the temperature. It also features a ventilation fan to manage humidity levels and an automated water pump to maintain the perfect soil moisture.

The design requirements set limits on the prototype volume, which should be less than 0.2 m³. The structure features a rectangular base with a 60 cm length, 37 cm width, and a 44 cm height, topped with a sawtooth roof to optimize light and ventilation. Beyond the physical build, the prototype's artificial intelligence must operate with at least ±80% accuracy to ensure reliable crop management.

Materials & Methods

Table 1 - List of Materials Used in Prototype

Item	Description	Quantity	Picture
Wood + Polyethylene Bubble Wrap	Main recycled parts used in prototype body construction.	4 Blanks (60*40cm)	
ESP32 + Power Supply	Main parts in the circuit for energy supply and control	1 Controller + 1 Supply	
Peltier Kit	Actuators used to regulate temperature.	2 Kits	
DHT22	A sensor reads humidity and temperature.	1 Module	
Soil Moisture Sensor	A sensor reads water percentage in soil.	1 Module	
Water Pump	Actuators used in irrigation and drainage system	2 Pumps	
Servo Motor + Fan	Actuators used in ventilation system for humidity regulation.	1 Servo + 1 Fan	

Methods

1- Creating a 3d model: a 3d design with the expected shape of the prototype was designed to be a guide during process of construction.

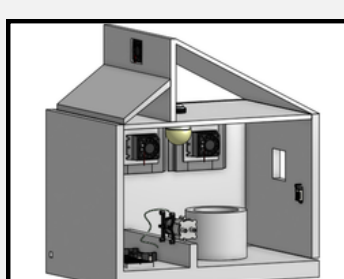


Figure 2 - 3d Design

2- Schematic diagram: a schematic diagram for the circuit was created to be a guide during soldering process and to code the firmware for the esp32.

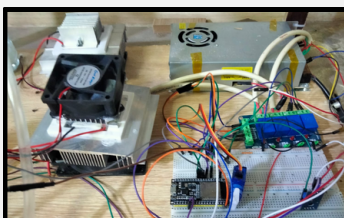


Figure 3 - Connected Circuit

3- Collecting materials: recycled wood was collected from an old desk and leftover blanks.



Figure 4 - Assembled Prototype

4- Wiring circuit and coding firmware: the circuit was soldered based on the schematic diagram and coded on Arduino IDE.

5- Constructing the body: the body was constructed using the recycled wood and shaped into a sawtooth roof and holes were cut perfectly for ventilation system.

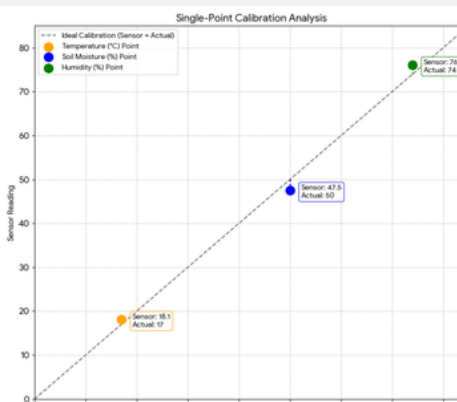


Figure 5 - Calibration Curve

6- Assembly and Calibration: After connecting the circuit, burning firmware, combining it with the body and sensor calibration, the prototype finally became ready for the test plan.

Test Plan

Sensor Calibration: Sensor readings were monitored continuously and calibrated. Temperature and humidity readings are calibrated with their value in the specific day weather, and soil moisture is calibrate with calculations.

Prototype Monitoring: The prototype was monitored for 6 continuous hours and checked for signal speed, actuators functionality, proper sensor readings and condition maintaining, these results.

Data Collection and AI Testing: The monitored data was collected and saved to an online database, and the model diagnoses the pictures of the plant for diseases.

Results

Negative Results

Due to the low memory size of esp32 and connecting series actuators to multiple pins instead of one, the overload had burned the controller. To overcome this, a new controller was used and each system was connected to a single pin using relays.

Because of the improper testing of the peltier without using heat sinks and fans, the first peltier had burned due to the significant temperature increase. To solve this issue, a new peltier was connected to its kit and installed in the prototype.

Positive Results

The total power consumption of the prototype didn't exceed 259 Watts considering the activation of all actuators and sensors.

The prototype succeeded to maintain the optimal condition of the crop for 5 consecutive days in specific times as shown in Figure 6,7 and 8.

After sensor calibration, the error percentage of temperature was ±2.1°C, the error of humidity was ±1.1% and soil moisture error ±3.6%.

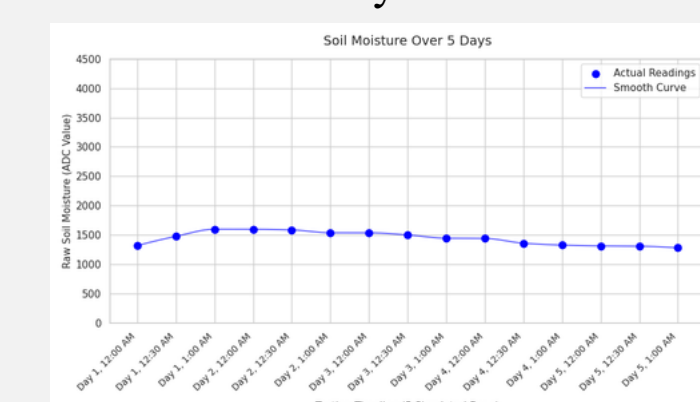


Figure 6 - Soil Moisture Measurement Trials

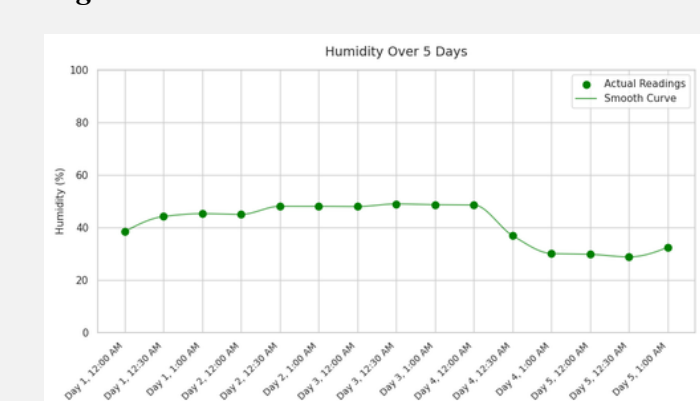


Figure 7 - Humidity Measurement Trials

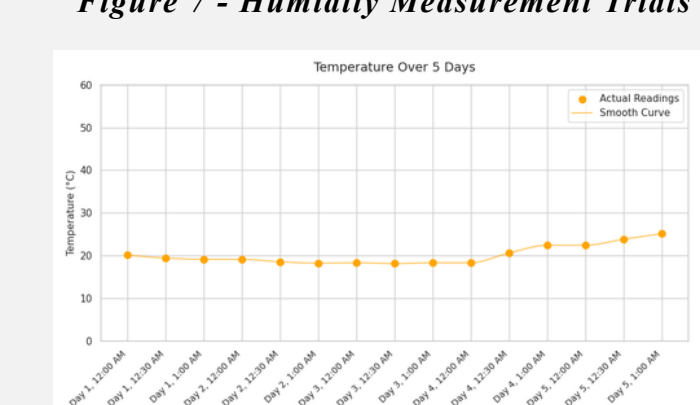


Figure 8 - Temperature Measurement Trials

Table 2 - Table of Measurements Over 5 Days

Day / Time	Room Soil Moisture (ADC)	Humidity (%)	Temperature (°C)
Day 1, 12:30 AM	1337	38.5	26.2
Day 1, 12:30 AM	1474	44.1	19.4
Day 1, 1:00 AM	1596	45.2	19.1
Day 2, 12:00 AM	1595	44.9	19.1
Day 2, 12:30 AM	1584	48.0	18.5
Day 2, 1:00 AM	1535	48.0	18.2
Day 3, 12:00 AM	1535	47.9	18.3
Day 3, 12:30 AM	1489	48.9	18.1
Day 3, 1:00 AM	1442	48.6	18.3
Day 4, 12:00 AM	1439	48.5	18.3
Day 4, 12:30 AM	1357	36.8	20.6
Day 4, 1:00 AM	1325	30.0	22.4
Day 5, 12:00 AM	1311	29.7	22.4
Day 5, 12:30 AM	1366	28.7	21.8
Day 5, 1:00 AM	1279	32.4	25.1

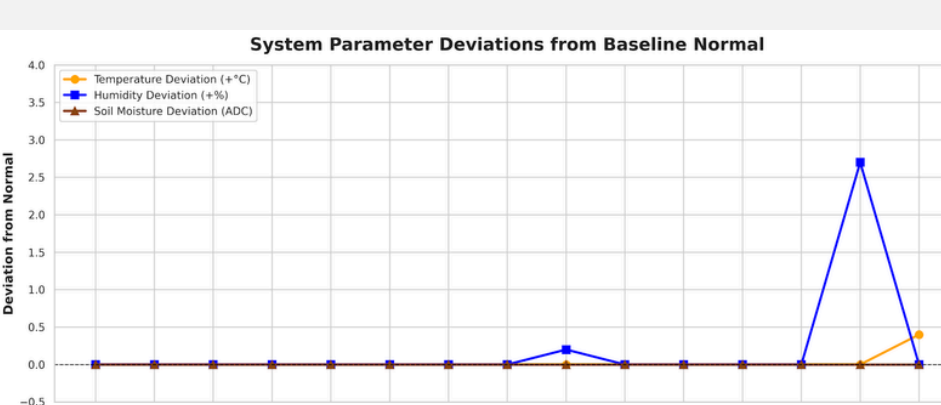


Figure 9 - Deviation Curves

Analysis

Heat Transfer

Thermoelectric Peltier module is one of the most efficient tools for cooling systems. They work by moving heat from the side to the other side when you run an electric current through them. The amount of cooling they can actually do is affected by a few things. Including heat getting in where it should not how well the insulation works and the difference in temperature between the two sides. But how well can a TEC-12706 work in a greenhouse model that is 0.125 m³ ? To determine the cooling power of the TEC-12706, Equation 1 was used:

$$Q_c = \alpha I T_c - \frac{1}{2} I^2 R - K(T_h - T_c)$$

Equation 1 - Peltier Cooling Power

Using values like current of 12 Amperes, voltage of 12 Volts, resistance of 1.0 Ohm and thermal conductivity of 1 Watt per Kelvin the cooling power of the system was about 72 Watts per module. The system had two modules so the total cooling power was 144 Watts. The thing is, polyethylene is not an ideal insulator so some of the cooling power is lost because of heat coming back and air escaping. If the insulation is really bad the system does not work well and the temperature only goes down by 2 to 4 degrees Celsius. If the insulation is medium the system works better and the temperature goes down by 6 to 8 degrees Celsius inside the greenhouse. To make the Peltier-based system work well it is required to make sure there are no air leaks the hot and cold sides are separate and the air is moving inside. This way the Peltier-based system keeps working in a way and controls the temperature well which is good, for small greenhouses (Chhanbta & Galphade,2025).

Roof Analysis

The sawtooth roof is an architectural design that use passive ventilation to naturally regulate the internal climate of a greenhouse. By allowing sunlight to enter while preventing the buildup of solar heat, this structure ensures a stable thermal environment. When paired with an active ventilation system which triggers only during periods of high humidity. Equation 2 is used to understand how air pressure changes as wind moves over a structure

$$C_p = \frac{(P - P_0)}{(0.5 \rho U_{ref}^2)}$$

Equation 2 - Pressure Coefficient

In this equation, P is the local static pressure on the roof, while P_0 is the surrounding atmospheric pressure. The variable ρ represents the air density, and U_{ref} is the speed of the wind hitting the structure. Together, these factors determine C_p which indicates whether a specific area has the low pressure needed to naturally pull out hot air (Perén et al.,2015).

Feedback System

The prototype uses a system to maintain a perfect condition for the crop called feedback system. This system consists of sensors and actuators that work whenever the environmental conditions changes and becomes hard for the plant to grow in. The process starts when a reading of a sensor exceeds the normal range defined in the firmware, the esp32 gives a signal to relay connected to a specific actuator to activate it until this parameter returns to normal range. Figure 10 shows the schematic diagram for actuators and sensor where signals are sent from digital pins in the esp to the relays. The actuation system consists of 3 parts, when the temperature is higher than 25°C, the esp32 sends a signal for the peltiers to work until temperature is back under 25. when soil moisture reaches 0%, the water pump activates until it reaches 20%, and when the humidity increases above 50% the controller sends a signal to the servo motor to open the ventilation and activate the top fan which lets out hot humid air.

Electric Power

Electric power is defined as the rate at which electrical energy is produced, transmitted, or consumed in a circuit. It indicates how quickly electrical energy is converted into other forms such as heat, light, or mechanical energy. Electric power is commonly measured in watts (W), and it can be calculated by multiplying voltage by current, as shown in Equation 3. $P = VI$

Equation 3 - Power Law

In the project, electric power is essential for operating the circuit electronics systems, as it determines how much energy a circuit uses over time. Efficient management of electric power is important in the project, where it helps ensure reliable and optimised performance.

Table 3 - Learning Outcomes00

Learning Outcome	Concept	Content	Relation
ES.2.09	3	Development of Crust	Helps to determine the best location and soil for crop
MA.2.07	1,3 and 6	Derivatives, Power Function	Helps to graph the data and determine power in a specific moment
ME.2.05	1	Power	Helps to calculate power consumption
CS.2.05	1 to 5	App Structure	Helps to determine the structure of an app to display data.
CH.2.08	2,5 and 8	Corrosion	Helps to define the best material to build the prototype

Conclusion

Egypt suffers from extreme temperatures, saline ground, and poor performance of conventional greenhouses, which increase costs and decrease sustainability. The aim of this project was to design a smart greenhouse to control climate, promote plant growth, and save resources. Lavender was selected as a test subject to assess the system's efficiency under challenging circumstances. The prototype combines the best features of previous systems with improvements in automation, structural optimization, and climate regulation.

Recommendations

Wadi El-Natroun

Wadi El-Natrun is one of the most capable areas to build a lavender based soil restoration prototype. The area is known for its very salty and alkaline soils, few plants, and bad weather, all of which make farming much less productive. These issues make it a great place to see how lavender can help restore land, improve soil structure, and deal with high salt levels. Applying this project in Wadi El-Natrun would not only show that it works in real life, but it could also help restore a lot of land in one of Egypt's most damaged semi arid areas such as Wadi El-Natrun (Yousif & Ahmed, 2019).

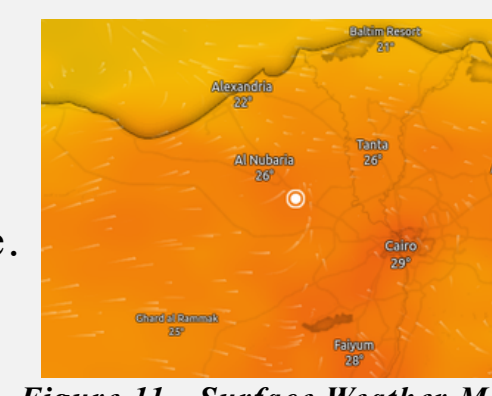


Figure 11 - Surface Weather Map (Windy.com, 2026)

Galvanized Steel

Although wood was used to construct current prototype for its recyclability, it has limitations in long term durability and resistance to environmental conditions. Moisture damages wood more easily, and it can deform and weaken structurally over time. For higher structural efficiency, it is recommended to use galvanized steel as an alternative material for full scale greenhouses. It delivers high strength and resistance to corrosion because of its zinc coating. However, it is more expensive than wood requiring specialized tools and labour (Castro et al.,2024)



Figure 12 - Galvanized Steel

Iron Low-Tempered Glass

Although polyethene was used in prototypes because it is cheap and easy to recycle, it does not work well for real projects. Polyethene lets in light, but it gets worn out over time when exposed to sunlight. To make the greenhouse more like real farming conditions, low-iron tempered glass is a good choice. This type of glass lets in a lot of light, 90 to 92%, which helps plants grow. It is also 4 to 5 times stronger than glass. However, it costs more than \$140 per meter in Egypt(Castro et al.,2024).

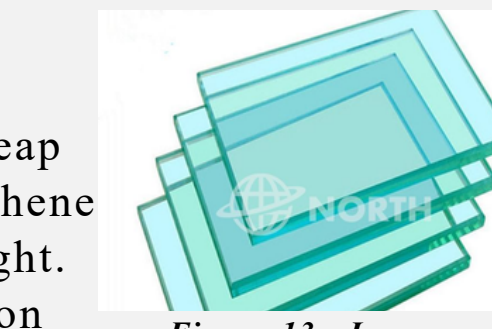


Figure 13 - Iron Low-Tempered Glass

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